

Reading the Latest, Not the Loudest: Lifecycle-Aware Structured Memory for Retrieval-Augmented Generation

Anonymous submission

Abstract

Retrieval-augmented generation feeds a language model the passages that look most relevant to a question, but relevance says nothing about whether a passage is still true. Real corpora describe the same fact many times as it changes over the years, so a question like “who is the CEO?” turns up both outdated and current answers. We introduce WIKI-GRAPH-RAG, a retrieval memory that tracks each fact’s *lifecycle*. From span text and source timestamps, without an LLM or training labels, it links a statement to a later changed statement of the same subject and relation. Before top- k selection, the retriever filters statements predicted to be superseded. We formalize the guarantee under exact key/value gates and valid temporal order, making matcher, value-comparison, and timestamp failures explicit. Across six datasets, detection is near-perfect on rendered structured claims and reliable on genuine Wikipedia prose; on matched Wikidata pairs it also outperforms a prompted LLM judge. Under an identical corpus, question, five-claim budget, prompt, reader, and scorer, lifecycle retrieval reaches 0.937–0.970 current-answer exact match (EM) on Wikidata and third-party TempLAMA, versus 0.121–0.260 for TempRALM and 0.473–0.697 for development-selected RAG-Time ($p < 0.001$ in all four aggregate comparisons). Grouping edge-connected claims and sorting them by timestamp additionally yields 0.940–1.000 historical top-1 accuracy, versus 0.388–0.750 for TempRALM and 0.873–0.972 for RAG-Time; the resulting intervals compose across hops. Every number is backed by a SHA-256-verified reproducibility bundle.

1 Introduction

Retrieval-augmented generation (RAG) grounds large language models in external text (Lewis et al. 2020; Izacard et al. 2022; Ram et al. 2023): a retriever returns the top- k passages by similarity to the query, and a reader conditions on them. This pipeline implicitly treats the corpus as a set of mutually consistent facts. Real corpora are not. They accrete over time, so a single entity attribute, such as a country’s head of government or a policy’s income threshold, appears in the corpus as a *sequence* of values, each true for a while and then replaced. Relevance is blind to this. The passage stating “the

CEO is X” and the later passage stating “the CEO is now Y” are about equally similar to *who is the CEO?*, so flat retrieval routinely packs both into the reader’s context, and the reader, given no recency signal, can answer with the stale one.

Existing remedies stop short of this problem, and we defer their detailed comparison to the next section. In brief, structured and graph retrievers reorganize a corpus into entities and relations but treat an attribute as an unordered bag of every value it ever held; temporal-QA benchmarks document the failure and search-augmented methods re-query the live web rather than equip a static-corpus retriever; and knowledge editing rewrites a model once an oracle has already decided which fact is current. The question we ask sits upstream of all of these: can a retriever decide, from the text alone, which of its own claims have been superseded, and use that to retrieve the current one?

Approach. We introduce WIKIGRAPH-RAG, a structured RAG memory in which the lifecycle of a claim is a retrieval signal (Fig. 1). The detector reads only span text and source timestamps, using no language-model call or gold key/value label: text-derived gates test shared subject and relation, changed value, and temporal order (Algorithm 1). At query time a lifecycle mask removes predicted-superseded claims before top- k context selection.

Why a hand-built rule instead of a model? We test a prompted gpt-4o-mini judge on matched Wikidata span pairs (Table 3); it reaches 0.960 F1 while the zero-LLM detector reaches 1.000. We then characterize the boundary of that result (Proposition 1, Lemma 1): exact key/value gates and valid temporal order give exact detection, while practical errors trace to those auditable inputs rather than a generated judgement.

Contributions. We (i) recast a claim going out of date as a text-and-time link and build an auditable detector with no language-model call or training labels; it reaches 0.976 F1 on published TempLAMA and outperforms a prompted LLM judge on the tested Wikidata pairs. We (ii) prove the which-value failure of ordinary retrieval and test it in a direct temporal-RAG head-to-head across two datasets and two readers. We (iii) use edge connectivity to group timelines, timestamp-order their validity intervals, and compose dated facts across hops. Finally, we (iv) release every headline result behind a checksummed bundle.

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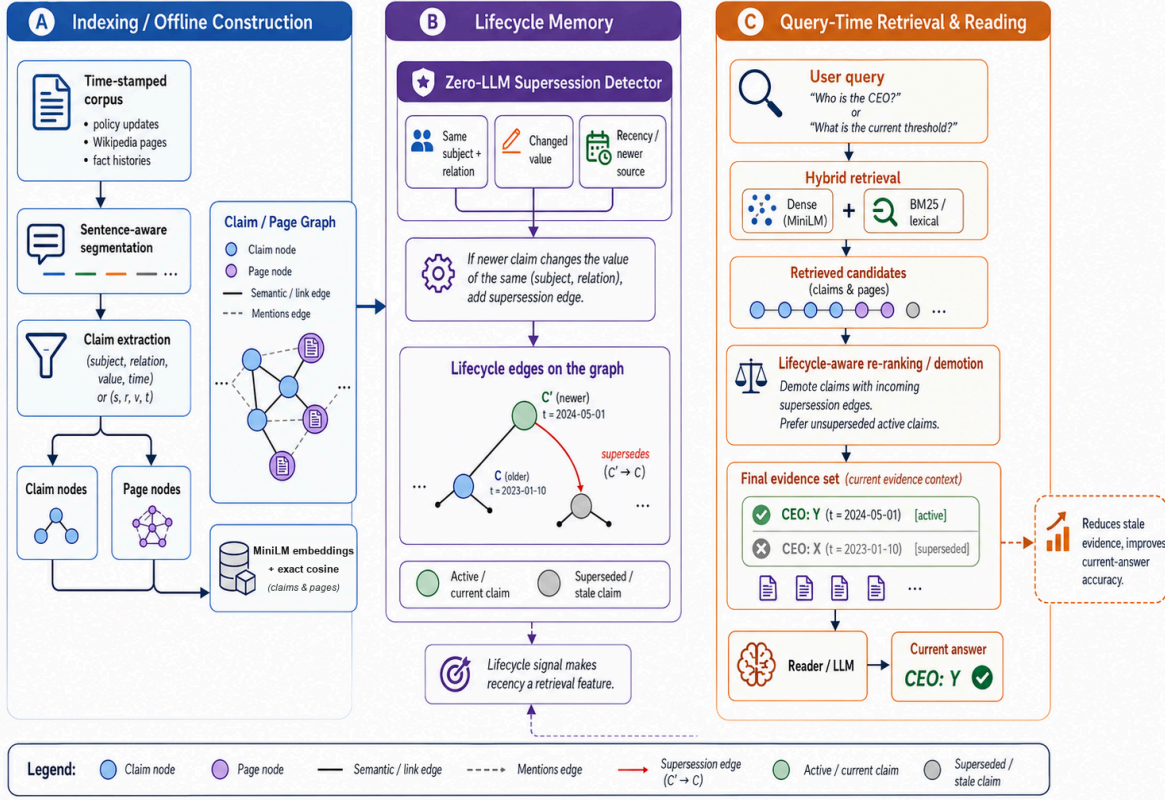


Figure 1: **WIKIGRAPHRAG overview.** (A) **Offline construction.** A time-stamped corpus is segmented into source-attributed claim spans; frame, value, and phrase features are extracted from each span, while claim and page nodes form a graph and are embedded with MiniLM for exact cosine scoring. (B) **Lifecycle memory.** The zero-LLM detector (Algorithm 1) adds an edge when a newer claim changes the value of the same subject and relation. (C) **Query-time retrieval.** Before top- k selection, a lifecycle mask filters every claim carrying an incoming supersession edge, so the evidence holds the active value (CEO: Y, dated 2024) instead of the superseded one (CEO: X, dated 2023). Edge-connected timelines are sorted by timestamp to support as-of and temporally consistent multi-hop queries without another model call.

2 Related Work

Structured and graph RAG. HippoRAG (Gutiérrez et al. 2024) builds an open-information-extraction graph and runs personalized PageRank (Haveliwalla 2002) seeded by query entities; GraphRAG (Edge et al. 2024) clusters an entity graph into communities and summarizes each with an LLM. Neither models time: a node accumulates every value an attribute ever held, with no notion that one value supersedes another. WIKIGRAPHRAG adds a typed supersession edge over a claim/page graph, so the structure that supports multi-hop retrieval also encodes which value is current. Tree-organized retrievers such as RAPTOR (Sarhi et al. 2024) and TreeRAG (Tao et al. 2025) are likewise lifecycle-agnostic.

Temporal and freshness RAG. TempLAMA (Dhingra et al. 2022), TimeQA (Chen, Wang, and Wang 2021), StreamingQA (Liška et al. 2022), and RealTime QA (Kasai et al. 2023) expose stale knowledge; FreshLLMs (Vu et al. 2024) instead re-queries the live web. TimeRAG decomposes complex temporal questions and iteratively couples web search with time-aware answer generation (Wang, Zhao, and Dou

2025). Timestamp-aware retrievers blend semantic relevance with inverse time distance (TempRALM) (Gade and Jetcheva 2024) or a half-life recency prior (Grofsky 2025); FRESKO shows that even strong rerankers remain biased toward obsolete passages (An et al. 2026). TA-RAG retrieves contiguous evidence over an extracted temporal window (Lau et al. 2025), whereas IA-RAG organizes interval events and applies interval-algebra constraints (Wang et al. 2026). Temporal KGs (Trivedi et al. 2017; Lacroix, Obozinski, and Usunier 2020) likewise operate on timestamped events or supplied validity intervals. Our task is upstream of search-time reasoning: infer supersession from timestamped claim spans, then derive intervals from inferred timeline groups. We evaluate on TempLAMA and directly compare with TempRALM and the half-life prior.

Knowledge conflicts and editing. Entity-based knowledge conflicts (Longpre et al. 2021) and studies of reader behavior under conflicting context (Xie et al. 2024; Liu et al. 2024) show that models are swayed by what is retrieved; model editing (De Cao, Aziz, and Titov 2021; Meng et al. 2022)

rewrites parameters once an oracle supplies the new fact. Our work is upstream of editing: it decides from text which claim is current and feeds that to the reader through retrieval rather than parameter surgery. Standard dense (Karpukhin et al. 2020; Reimers and Gurevych 2019) and lexical (Robertson et al. 1995) retrieval are used as-is.

3 Method

Problem setup. The memory receives source-attributed claim spans x_c with sortable timestamps t_c . For analysis, associate each span with $c = (s_c, r_c, v_c, t_c)$: subject, relation, value, and source (or stated-validity) time. These tuple fields define evaluation gold but are never read by the detector or retriever, which use only x_c and t_c . Write the latent subject-relation key $\kappa(c) = (s_c, r_c)$ and $\mathcal{C}_\kappa = \{c : \kappa(c) = \kappa\}$. We study functional keys with one value in force at a time; concurrent and non-substitutive relations are outside scope. A claim is superseded when a later same-key span carries a different value:

$$S^* = \{c \in \mathcal{C} : \exists c' \in \mathcal{C}_{\kappa(c)}, t_{c'} > t_c, v_{c'} \neq v_c\}, \quad (1)$$

and the active set is $A = \mathcal{C} \setminus S^*$. Claim text remains verbatim rather than generated; span selection, text-derived gates, and temporal metadata can nevertheless be wrong and are evaluated separately. Claim and page nodes are embedded with all-MiniLM-L6-v2 (Reimers and Gurevych 2019); the implementation scores exact cosine similarity over the embedding matrix.

Text-driven supersession detection. The detector estimates the relation in Eq. (1) from text alone, with no language-model call and no gold label. It needs three signals. (1) *Subject and relation*: each claim’s tokens split into a *frame* (relation words, e.g. “head of government of”) and *value objects* (numbers or proper-noun spans); two claims share a relation when frames overlap by an overlap coefficient of at least 0.5, and share a subject when, given that both name at least two proper nouns, they share a full capitalized phrase (e.g. “United Kingdom”), falling back to strong token overlap when a claim names at most one proper noun (then that noun is most likely the changing value). Phrase-level matching is what stops two subjects sharing only a generic suffix (“... Liberal Party” versus “... Republican Party”) from cross-linking, while still matching one subject across its value changes. (2) *Changed value*: comparing only like-typed claims, a numeric claim changes when its unit-normalized magnitude differs (ignoring bare years and identifiers), and a string-valued claim changes when each side carries a content token the other lacks, i.e. a substitution rather than a restatement. (3) *Recency*: the edge is oriented from the later source to the earlier one, and lexical cues (“revised”, “replaces”, “with effect from”) raise edge confidence from 0.7 to 0.9 but are not required. Algorithm 1 keeps, for each claim, the single newest claim that supersedes it. The detector is worst-case quadratic in claims, but the gates reject most pairs at the first comparison; the largest combined corpus (712 claims) adds 0.3 CPU-seconds at indexing time.

The detector has two heuristic gates: subject-relation matching and changed-value comparison. We record a diagnostic boundary, not an unconditional guarantee. Let $M(a, b)$

Algorithm 1 Text-driven supersession detection

Require: claims $\{c_i\}$ with text and source timestamps t_i
Ensure: edges E ; each (c', c) means “ c' supersedes c ”

```

1: for each claim  $c_i$  do
2:    $(f_i, v_i, p_i) \leftarrow (\text{FRAME}, \text{VALUES}, \text{PHRASES})(c_i)$ 
3: end for
4:  $E \leftarrow \emptyset$ 
5: for each candidate stale claim  $c$  do
6:    $best \leftarrow \text{NONE}$ 
7:   for each  $c' \neq c$  with  $t_{c'} > t_c$  do
8:     if not SAMESUBJECTRELATION( $c, c'$ ) then continue
9:     end if
10:    if not CHANGEDVALUE( $c, c'$ ) then continue
11:    end if
12:    if  $best = \text{NONE}$  or  $t_{c'} > t_{best}$  then  $best \leftarrow c'$ 
13:    end if
14:  end for
15:  if  $best \neq \text{NONE}$  then  $E \leftarrow E \cup \{(best, c)\}$ 
16:  end if
17: end for
18: return  $E$ 
```

and $V(a, b)$ denote these predicates, and let $\hat{S}$ be the predicted superseded set.

Proposition 1 (Conditional exactness). *If $M(a, b) \Leftrightarrow \kappa(a) = \kappa(b)$, $V(a, b) \Leftrightarrow v_a \neq v_b$ on comparable claims, and timestamps follow validity order, then $\hat{S} = S^*$.*

Proof. The detector marks c iff some later c' satisfies $M(c', c)$ and $V(c', c)$; the premises make this identical to membership in S^* (Eq. (1)). $\square$

The statement is near-immediate, and that is its point: it localizes error to M , V , or temporal metadata rather than certifying practical gates. Observed prose errors below are dominated by inspectable key merges and splits; value and timestamp perturbations are evaluated separately.

Lifecycle-aware retrieval. Hybrid dense and lexical scoring produces an ordered pool π_q . Before truncation, the lifecycle layer hard-masks every claim with an incoming predicted-supersession edge: $R_q = \text{Top}_k\langle c \in \pi_q : c \notin \hat{S} \rangle$. Masked claims are not appended later; with five same-key values and three slots, an exact mask admits the one active value, not two stale fillers. We report stale-evidence rate and active-evidence precision,

$$\text{SER} = \Pr_q[R_q \cap S^* \neq \emptyset], \quad \text{AEP} = \mathbb{E}_q \left[\frac{|R_q \cap A|}{|R_q|} \right]. \quad (2)$$

The next lemma isolates the failure as one of *which value*, not recall.

Lemma 1 (Which-value). *Condition on a same-key conflict block of $m > 1$ claims (one active) and $1 \leq j \leq m$ context slots. If relevance is value-agnostic within the block and ties are uniform, flat retrieval includes the active claim with probability j/m , admits stale evidence surely when $j \geq 2$*

(and with probability $(m - 1)/m$ when $j = 1$), and has expected block-level AEP $1/m$. With the exact mask S^* applied before truncation, lifecycle retrieval retains the active claim, filters all $m - 1$ stale claims, and has SER 0 and AEP 1 on the block.

Proof. Exchangeability makes the j admitted a uniform j -subset, giving the flat quantities directly. Exact filtering leaves the sole active claim as the only admissible member of the block. $\square$

Lemma 1 predicts the pattern: flat retrieval leaves stale claims co-resident, while exact filtering sends $SER \rightarrow 0$ and $AEP \rightarrow 1$. The next bound concerns only interventions applied after a context has been selected.

Proposition 2 (Fixed-context ceiling). *Let flat retrieval first select a uniform j -subset R from a same-key block of m claims, where $j < m$. Any post-selection method that only reorders or annotates R leaves active-claim inclusion at j/m . If the reader errs with probability at least $\delta > 0$ when the active claim is absent, expected exact match is at most $1 - \delta(m - j)/m$.*

Proof. Reordering or annotation preserves fixed R . By Lemma 1, the active claim is absent with probability $(m - j)/m$; conditioning on that event yields the bound. $\square$

This does not cover a full-pool temporal reranker that changes membership before top- k ; we evaluate TempRALM and RAG-Time directly. The history sweep separately tests post-selection date prompting and newest-first ordering (Table 5).

Validity intervals and as-of retrieval. Algorithm 1’s edge need not point to the immediate next change; here it groups edge-connected claims into an inferred timeline. Within each component, sorting distinct values by timestamp gives adjacent change points $c_1, \dots, c_m$ and intervals $[t_i, t_{i+1})$, with the final interval ending at ∞ . An as-of query returns $\arg \max_{c: t_c \leq T} t_c$ from the matched component. Thus timestamps define boundaries; edges define text-derived timeline membership.

Proposition 3 (As-of correctness). *If edge-connected components equal the true key groups and timestamps order their true change points, the as-of operator returns the value valid at every probed T .*

Proof. Within the correct component, adjacent timestamp order gives the true intervals, and the latest start not after T is exactly the value in force. Errors can arise from a wrong component or temporal order. $\square$

4 Experimental Setup

Detection benchmarks and provenance. FreshRAG is synthetic; Real-world is hand-curated; Wikidata labels and TempLAMA records supply structured subject, relation, value, and time fields that are rendered into claim text; Wikidata-prose paraphrases those same records. These annotations define gold only: the detector receives rendered text and time, never tuple fields. Real-Wikipedia contains 26

Input family	Visible to detector	Gold source (scoring only)
Synth/curated	rendered span + source date	generated or curated tuple
Structured	rendered span + benchmark time	Wikidata/TempLAMA record
Paraphrased	rewritten span + inherited time	inherited Wikidata tuple
Genuine prose	Wikipedia lead + tenure start	curated chronology order

Table 1: **Detector input versus evaluation gold.** Tuple fields construct benchmarks and labels but are not read by Algorithm 1. The full dataset-by-dataset map is Supplementary Table S1.

genuine CEO lead extracts across 12 chronologies, and RealProse adds 132 genuine leads over five relation families; these are the raw-prose parsing tests. Dataset sizes and label construction are detailed in Supplementary Secs. C–D.

This separation matters for interpretation: the four rendered benchmarks isolate supersession decisions once a claim span exists, whereas Real-Wikipedia and RealProse jointly test whether the same text-derived gates survive genuine lead prose, aliases, appositives, and pooled cross-subject hard negatives. We therefore report the two evidence tiers separately rather than treating six dataset names as six equivalent raw-text tests.

Metrics and statistics. Detection reports span-level precision, recall, and F1 of superseded-claim identification with false positives; retrieval reports SER and AEP (Eq. (2)); QA reports exact match (EM), the normalized gold value appearing in the answer. Confidence intervals are 95% cluster bootstraps over cases (2,000 resamples); supersession and QA gains use paired bootstrap tests, with a 5-seed mean where a condition is randomized. Every headline EM margin survives a strict clean match that rejects an answer also naming a stale value (Supplementary Sec. G, “Open-weights reader and strict scoring”).

Readers and baselines. Current-answer readers are deterministic extractive and local Qwen2.5-3B-Instruct (fp16, greedy); gpt-4o-mini is used only for the judge comparison. Every current-answer method shares corpus, question, $k=5$, prompt, reader, and EM scoring. TempRALM (Gade and Jetcheva 2024) and RAG-Time (Grofsky 2025) score the full pool from the same dense cosine similarities. RAG-Time’s $(\alpha, h)=(0.7, 1825 \text{ days})$ is selected from 20 settings on 9 Wikidata development keys by mean current/as-of accuracy, then frozen. The 33-key Wikidata aggregate includes those development keys; TempLAMA is fully external.

5 Results

Supersession is detectable from text. Table 2 reports span-level detection across all six benchmarks. FreshRAG, Real-world, Wikidata, and Wikidata-prose are exact (precision, recall, and F1 all 1.000). On TempLAMA, the detector reaches F1 0.976 (95% CI [0.964, 0.986]), with precision 0.982 and

Benchmark	Cases	P	R	F1	95% CI
FreshRAG	100	1.000	1.000	1.000	[1.000, 1.000]
Real-world	20	1.000	1.000	1.000	[1.000, 1.000]
Wikidata	73	1.000	1.000	1.000	[1.000, 1.000]
TempLAMA	300	0.982	0.971	0.976	[0.964, 0.986]
Wikidata-prose	73	1.000	1.000	1.000	[1.000, 1.000]
Real-Wikipedia	12	1.000	0.786	0.880	[0.667, 1.000]

Table 2: Text-driven supersession detection (single non-positional detector, Algorithm 1). P/R/F1 are span-level; CIs are cluster bootstraps over cases. Every value is regenerated from `results/detection_*.json`.

Benchmark Method	P	R	F1
Wikidata Heuristic (ours)	1.000	1.000	1.000
LLM judge (gpt-4o-mini)	0.923	1.000	0.960

Table 3: The zero-LLM heuristic vs a prompted gpt-4o-mini judge on the identical Wikidata span pairs (240 pairs, 120 positive), at no inference cost. Source: the released Wikidata judge card.

recall 0.971. On genuine Real-Wikipedia leads it reaches F1 0.880 at precision 1.000 and recall 0.786, localizing the degradation to missed links in raw prose rather than false supersessions. This is the pattern predicted by Proposition 1: crisp keys yield exact detection, while aliases and biographical clauses can split a timeline. Per-case ceilings alone do not certify the detector, so the pooled analysis below co-locates all subjects and contrasts the heuristic with newest-document-wins.

Matched-pair LLM comparison. Table 3 runs a prompted gpt-4o-mini judge on the identical 240 Wikidata span pairs. The zero-LLM detector scores 1.000 F1 versus 0.960: both find all 120 supersessions, but the judge adds ten false links (precision 0.923) while the structural rule adds none. This comparison is limited to the clean Wikidata pair set; Real-Wikipedia and RealProse evaluate the rule, but not an LLM judge, on genuine prose.

Lifecycle retrieval removes stale evidence. Table 6 isolates the retrieval effect and matches Lemma 1. Without lifecycle, every current-fact context contains stale evidence on FreshRAG, Wikidata, and TempLAMA (SER 1.000), while AEP is only 0.408, 0.297, and 0.438. Lifecycle retrieval reduces SER to 0.000, 0.000, and 0.060 and raises AEP to 1.000, 1.000, and 0.988, respectively, essentially matching the gold-edge oracle. The retriever still targets the right topic in the flat condition; the failure is choosing among co-retrieved values, exactly the which-value failure of Lemma 1.

Direct temporal-RAG head-to-head. Table 7 changes only retrieval. On Wikidata, lifecycle reaches 0.970 with both readers, versus 0.121/0.152 for TempRALM and 0.697/0.546 for RAG-Time (extractive/Qwen). On all 300 TempLAMA questions it reaches 0.953/0.937, versus 0.213/0.260 for TempRALM and 0.657/0.473 for RAG-Time. Against development-selected RAG-Time, aggregate paired gains are 0.273 (95% CI [0.121, 0.424]),

Diagnostic	Wikidata		TempLAMA	
	Flat	TempR	Flat	TempR
Same-key claims in top 5	4.000	1.121	2.460	0.643
Active claim in top 5	0.879	0.273	0.993	0.350
Top 1 has requested key	1.000	0.121	0.997	0.287

Table 4: **TempRALM topicality audit.** Reciprocal global recency can promote recent off-key claims. Values are means or rates over the exact 33/300 headline questions. Source: `results/tempralm_topicality_audit.json`.

	$m=2$	$m=3$	$m=4$	$m=5$
n keys	33	29	22	15
Flat AEP (pred. $1/m$)	0.500	0.333	0.250	0.200
Flat EM	0.879	0.552	0.500	0.533
Date-prompt EM	0.909	0.655	0.409	0.600
Date-rerank EM	0.212	0.276	0.318	0.333
Lifecycle EM	1.000	1.000	1.000	1.000

Table 5: **History length isolates the mechanism** (Qwen reader; Wikidata chronologies truncated to their latest m values). Flat AEP matches the $1/m$ prediction of Lemma 1; date prompting and reranking retain stale values, while lifecycle holds the EM ceiling. Source: `results/history_sweep.json`.

0.424 [0.242, 0.606], 0.297 [0.237, 0.357], and 0.463 [0.407, 0.520]; all $p < 0.001$. On the disjoint 24-key Wikidata top-1 check, the gap is 0.125 and not significant ($p=0.222$). TempRALM’s reciprocal-time term often displaces topical evidence (Table 4).

Baseline safeguards. Nine Wikidata development keys (31 historical probes) select one of 20 RAG-Time settings; the selected setting is frozen for the other 24 keys and external datasets. The main Wikidata aggregate retains all 33 keys for paired comparability and is therefore labeled development-selected, not held-out. TempRALM and RAG-Time rescore the full pool, so neither is constrained by Proposition 2. Newest-first direction is unit-tested and reaches 1.000 with the deterministic first-item reader; its low Qwen score reflects position sensitivity when stale values remain.

History length isolates the mechanism, not a dataset quirk. Table 5 truncates the same Wikidata histories to m values. Flat AEP follows $1/m$, while lifecycle filtering stays at 1.000 EM. Newest-first ordering is intentionally post-selection: it retains every stale value and, with Qwen, exposes a position bias (0.212 EM at $m=2$); the deterministic first-item reader scores 1.000, confirming that sort direction is correct. On TempLAMA’s complete 31-chronology long-history slice, Qwen reaches 1.000 with lifecycle versus 0.710 flat, 0.645 with a latest-date instruction, and 0.290 newest-first (all $p < 0.001$). On 35 genuine Wikipedia-lead questions, lifecycle raises Qwen from 0.714 to 0.857 EM (gold 0.914) while SER falls from 0.971 to 0.229 (Supplementary Secs. E–F).

The inferred groups answer the whole timeline. Edge connectivity groups claims; timestamp sorting supplies adjacent

	FreshRAG		Wikidata		TempLAMA	
	SER	AEP	SER	AEP	SER	AEP
Flat RAG	1.000	0.408	1.000	0.297	1.000	0.438
WIKIGRAPHRAG	0.000	1.000	0.000	1.000	0.060	0.988
Gold (oracle)	0.000	1.000	0.000	1.000	0.000	1.000

Table 6: Lifecycle-aware retrieval drives stale-evidence rate (SER, lower better) to ≈ 0 and lifts active-evidence precision (AEP, higher better) to the oracle, with no gold labels at inference. Source: `results/retrieval_*.json`.

Method	Wikidata ($n=33$)		TempLAMA ($n=300$)	
	Ext.	Qwen	Ext.	Qwen
Flat RAG	0.182	0.424	0.453	0.567
TempRALM	0.121	0.152	0.213	0.260
RAG-Time	0.697	0.546	0.657	0.473
WIKIGRAPHRAG	0.970	0.970	0.953	0.937
Gold oracle	0.970	0.970	0.997	0.970

Table 7: Current-answer exact match (EM) under a direct head-to-head: each method receives the identical corpus, question, five-claim budget, prompt, reader, and scoring. TempRALM and RAG-Time alter only retrieval; RAG-Time uses the development-selected $(\alpha, h)=(0.7, 1825 \text{ days})$ setting. Source: the four released `results/qa*` cards.

validity boundaries (Proposition 3). TempRALM reaches 0.388–0.750 historical top-1 accuracy, development-selected RAG-Time 0.873–0.972, and lifecycle intervals 0.940–1.000 (Table 8). Lifecycle exceeds RAG-Time by 0.067 on Wikidata (95% CI $[0.000, 0.134]$, $p=0.055$), 0.063 on fully external TempLAMA ($[0.034, 0.092]$, $p < 0.001$), and 0.028 on Real-world ($[0.000, 0.083]$, $p=0.739$); only TempLAMA is significant. Against TempRALM, gains remain 0.552, 0.302, and 0.250 (all $p < 0.001$).

The intervals compose across hops. Chaining two validity intervals answers a temporally-consistent multi-hop query flat and current-only retrieval cannot. On real Wikidata chains of 10 genuine subsidiary–parent–CEO timelines with real acquisition and tenure dates (*e.g.* Whole Foods $\rightarrow$ Amazon $\rightarrow$ Bezos, Red Hat $\rightarrow$ IBM $\rightarrow$ Rometty), resolving both hops as-of the query time answers 1.000, matching the gold-interval oracle, whereas a current-value graph chaining latest-to-latest answers only 0.200: every answer is present but assembled from mismatched eras. A controlled 40-query two-hop benchmark shows the same (1.000 vs. 0.000; flat surfaces both bridging claims for only 0.100, as the intermediate entity never appears in the query). Composition inherits its exactness from detection (Proposition 3).

6 Analysis

The three signals play distinct roles. Table 9 lesions each detector signal. Recency is the backbone: removing reliable order reduces F1 from 1.000 to 0.638 on FreshRAG and from 0.976 to 0.600 on TempLAMA, because an edge cannot be oriented without it. Approximate order is sufficient: replacing one quarter of dates by wrong in-range

Benchmark	TempR	R-Time	exact-KG	Ours
Wikidata	0.388	0.873	0.925	0.940
TempLAMA	0.671	0.911	0.933	0.973
Real-world	0.750	0.972	1.000	1.000

Table 8: **Historical as-of accuracy** under identical probes and pools. TempRALM and development-selected RAG-Time score the full pool; exact-KG and lifecycle group time-lines and sort by time. Probe counts: 134/447/36.

Benchmark	Full	–subject	–value	–recency	–all
FreshRAG	1.000	0.759	0.940	0.638	0.737
Real-world	1.000	0.911	1.000	0.667	0.725
Wikidata	1.000	0.887	1.000	0.752	0.774
TempLAMA	0.976	0.778	0.986	0.600	0.733
Wikidata-prose	1.000	0.887	1.000	0.752	0.774

Table 9: Ablation of the detector’s three signals (span-level F1). Recency is the backbone everywhere; subject matching is essential on multi-entity corpora, while the value gate mainly guards FreshRAG. Randomised conditions (–recency, –all) average 5 seeds. Source: `results/ablation_*.json`.

years costs 0.060/0.113 F1 on Wikidata/TempLAMA (Table 10), whereas making the same fraction missing costs 0.185/0.234. Subject matching prevents cross-entity links: removing it lowers F1 to 0.759 on FreshRAG and 0.887 on Wikidata-prose. The changed-value gate is dataset-specific: removing it lowers FreshRAG from 1.000 to 0.940, but is slack or slightly conservative elsewhere (TempLAMA rises to 0.986 without it). Thus recency is universally necessary, subject matching matters under multi-entity noise, and the value gate protects corpora containing restatements.

The honest limit on raw prose. On 26 genuine Real-Wikipedia CEO leads, the detector reaches F1 0.880 at precision 1.000 and recall 0.786 (Table 2): aliases and long biographical clauses split some timelines, but produce no false supersessions. The broader RealProse benchmark tests 132 genuine leads across five relation families in one pooled hard-negative index. With the same zero-LLM parser rules it reaches F1 0.941 at precision 0.978 and recall 0.907 (Table 11), leaving every structured score unchanged. The residual misses are precisely the matcher failures Proposition 1 predicts.

Residual errors are inspectable. RealProse contains nine false negatives and two false positives. Four of five relation families retain precision 1.000; the only false merges are two football managers whose leads share “Premier League” while one omits the club from its head sentence. The false splits are likewise concrete: corporate aliases (“Twitter” versus “X Corp”, “Starbucks” versus “Starbucks Coffee Company”), relation frames diluted by long biographies (a FIFA president introduced as lawyer, businessman, and athlete; a World Bank president as physician and anthropologist), and leads that append a second office with “and”. Company chief executives and organization leaders account for six of the nine

Benchmark	dates	0	10%	25%	50%	100%
Wikidata	wrong	1.000	0.968	0.940	0.876	0.798
	missing	1.000	0.936	0.815	0.556	0.000
TempLAMA	wrong	0.976	0.933	0.864	0.765	0.595
	missing	0.976	0.890	0.742	0.453	0.000

Table 10: **Timestamp-noise robustness** (detection F1, mean of 5 seeds) as a fraction of source dates is replaced by a *wrong* in-range year or made *missing*, with gold held fixed. Wrong-but-present dates degrade the detector gently; only missing dates hurt sharply, and only in the limit.

Relation family	n	P	R	F1
Company chief executive	26	1.000	0.786	0.880
National head of government	38	1.000	0.933	0.966
National head of state	20	1.000	0.938	0.968
Football-club manager	19	0.882	1.000	0.938
International-org. leader	29	1.000	0.864	0.927
Pooled	132	0.978	0.907	0.941

Table 11: **Generalization on genuine prose** (RealProse: 132 real Wikipedia leads, five relation families, 35 subjects, one pooled index, no per-relation tuning). Precision is 1.000 on four of five families (0.978 pooled): the pooled hard negatives (eight “prime minister” timelines, president-of-US vs. president-of-FIFA) are not cross-linked.

misses, exactly where aliases and long appositives are common. Thus every observed error has the form predicted by Proposition 1: an erroneous merge or split of subject–relation keys, not an untraceable generated judgement. Entity aliasing and better clause segmentation are direct repair targets.

Pooling separates structure from recency alone. Across the four structured corpora pooled into one 1,315-claim index, detection reaches 0.985 F1 ($P=0.986$, $R=0.984$), close to 0.991 when cases are isolated; newest-document-wins instead falls from 0.985 to 0.756 because it cannot distinguish stable facts belonging to other subjects. The scaling replay below tests the same pooled edge decision under unique-subject distractors; construction remains offline, and query-time retrieval adds an edge lookup and mask.

“Current” is an evidence-state claim, not a truth guarantee. The method assumes that source timestamps order the assertions in the supplied corpus and that, within a subject–relation key, a later changed value replaces the earlier value. It therefore identifies the latest supported assertion in that corpus; it does not independently verify that the newest source is factually correct or more authoritative. A newer false report can consequently be marked active, while disagreement among simultaneous sources requires an authority or corroboration model outside our scope. Reversions are representable: if a value changes from A to B and later back to A, the final A claim becomes active and the intervening B claim is superseded. This distinction is why we evaluate edge precision, timestamp corruption, and genuine prose separately. Lifecycle memory manages which source-attributed assertion is in force; fact verification remains complementary.

7 Conclusion

Relevance tells a retriever what a passage is about, not whether it is still in force. WIKIGRAPHRAG adds a text-and-time supersession edge and filters predicted-stale claims before context selection. Its formal boundary makes key matching, value comparison, and temporal order explicit; the experiments show strong detection on rendered structured claims, useful transfer to genuine prose, and 0.937–0.970 current-answer EM against directly implemented temporal scorers. Edge connectivity groups claims into timelines, and timestamp sorting supports value-at-time- T retrieval without another model call.

Limitations

The zero-LLM detector is exact on structured or lightly paraphrased claims but reaches F1 0.880 on genuine Real-Wikipedia CEO leads, where aliases and clauses blur subject–relation keys. The broader RealProse evaluation reaches 0.941, but it is a separate pooled benchmark rather than a direct repair of those same leads. Primary QA uses rendered functional chronologies; genuine-prose QA has 35 questions, and real multi-hop evaluation has 10 chains, so this is initial evidence rather than a large uncured deployment. On TempLAMA, missed edges leave lifecycle 0.033–0.043 below the gold oracle, so better entity canonicalization remains useful. FreshRAG-Bench measures evidence quality (SER, AEP), not reader EM, because every variant fits the answer in context. The method also assumes at least approximate timestamps: wrong dates degrade gracefully, but missing dates eventually remove the ordering signal (Table 10). The formal model further assumes one value in force per key and strict temporal order. Co-holders, memberships, gradual change, and simultaneous disagreement should not be collapsed into supersession; equal-time claims remain unresolved. A publication date can also lag the event it reports. Such cases require relation cardinality, stated-validity extraction, or source authority and corroboration beyond the current rule. Finally, all benchmarks are English with crisp entity attributes.

Ethics Statement

We use public datasets (TempLAMA and Wikidata), a local open Qwen reader with greedy decoding, and an OpenAI model only for the explicitly reported judge comparison. Returning the current claim verbatim may reduce confident answers from outdated facts; a wrong edge could suppress a still-valid claim. Because masking is exclusionary, false-positive edges matter more than under soft reranking; genuine-prose precision therefore remains the relevant safety caveat. AI assisted with code and prose under author review.

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